

[ANNOTATED]

MEMORIZATION OR INTERPOLATION? WHAT PERTURBATION SENSITIVITY ACTUALLY DETECTS IN LANGUAGE MODELS

Anonymous authors

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ABSTRACT

Deciding whether a language model *memorized* a text, rather than producing it by generalization, matters for privacy, copyright, and contamination auditing. A recurring proposal is the **Perturbation Sensitivity Hypothesis (PSH)**: memorized data points should be unusually fragile to small input perturbations, because a memorized completion is anchored to the exact surface form of its prefix. We subject PSH to a controlled audit against *ground-truth pretraining memorization* — EleutherAI’s per-sequence records of which Pile sequences each pretrained Pythia model reproduces verbatim — using threshold-free AUC-ROC against in-distribution negatives, bootstrap intervals, and paired significance tests on identical sequences.

The hypothesis survives as a phenomenon; whether it yields a competitive detector depends on implementation. Perturbation sensitivity is a genuine graded signal (Spearman $\rho=0.565$ against the memorization score), yet the original PEARL specification is handicapped by a too-narrow intensity schedule: on identical sequences, (i) a perturbation-free self-consistency baseline (needing neither reference nor logits) matches or beats PEARL at all five model sizes we measure, significantly at two (e.g. 0.939 vs. 0.906 on Pythia-410M, paired $p=0.010$); (ii) widening the intensity grid $4\times$ lifts PEARL to 0.933 and statistical parity with that baseline ($p=0.67$) — so the underperformance is partly a fixable spec artifact, though parity comes at $3.6\times$ the query cost; (iii) roughly half the signal (0.12–0.24 AUC) is a “room-to-fall” confound — memorized sequences begin with more headroom to degrade — which controlling for NCD@0 removes; and (iv) fragility is **sparse** (only 7–11 of 32 prefix positions carry damage), reconciling PSH with reports that memorized text is on average *harder* to corrupt, but also explaining why intensity aggregation wastes most of its budget. At the measured base rates (0.18%–0.94%, rarer than usually assumed), black-box precision at 1% FPR is 0.096–0.318. We conclude that PSH is sound as a phenomenon but unpromising as a detector, and that reported gains for perturbation-based detection should be checked against a headroom control and a perturbation-free baseline before being believed.

What we changed: **REVISED** The old abstract claimed AUC 0.934–0.973 and framed the result as a clean validation of the hypothesis. Every headline number here is re-measured (the old ones did not reproduce), the comparison is now against perturbation-free baselines on identical sequences with paired tests, and the two confounds we found are stated up front.

How to read this annotated copy. **NEW** and **REVISED** mark headings that are new or rewritten. **Yellow shading** marks changed or added text. Each white box like this one says, in one or two plain sentences, what changed and why. A summary of every change appears on the page before the references. Anything unshaded is unchanged from the earlier draft.

1 INTRODUCTION

Large language models (LLMs) can *memorize* portions of their training data and reproduce them verbatim (Carlini et al., 2021; 2023). Detecting memorization matters for privacy (leaking personal data), copyright and legal compliance (e.g. the New York Times lawsuit against OpenAI), and for auditing benchmark contamination, where a test set leaked into pretraining inflates reported accuracy. The central difficulty is distinguishing **true memorization** — the model regurgitating a stored training sequence — from **interpolation** — the model producing correct output through genuine generalization. Both look identical from a single correct completion.

A further constraint is that the most important models are closed: their weights, logits, and training corpora are inaccessible. Most memorization and membership-inference tools assume access to per-token loss or to (parts of) the training data (Carlini et al., 2021; Shi et al., 2024; Nasr et al., 2023), which does not hold for commercial APIs. This motivates *black-box* detectors that operate purely on generated text.

The hypothesis. We study the **Perturbation Sensitivity Hypothesis (PSH)**:

For a given model, task, and data point, if the model has memorized that data point, its task performance exhibits high sensitivity to small input perturbations.

The intuition: a memorized completion is bound to the exact stored key; nudging the input off that key removes the retrieval cue and performance collapses, whereas a genuinely generalized completion degrades gracefully. We implement PSH as **PEARL** (Perturbation Analysis for Revealing Language-model memorization): perturb an input at increasing intensities, measure how sharply the greedy completion diverges from the reference (via normalized compression distance), and flag a large falloff.

Provenance. PSH and PEARL were introduced in an earlier version of this work whose validation had four defects: memorization was *induced* by fine-tuning Pythia for ten epochs (overfitting, not pretraining memorization); detection was reported as hand-tuned counts with no ground truth; negatives were drawn from RefinedWeb, which shares Common Crawl ancestry with the Pile and so confounded “non-memorized” with “out-of-distribution”; and closed-model claims rested on 100 GPT-4o samples with speculative labels. We replace that validation entirely: natural pretraining memorization, published per-sequence ground truth, in-distribution negatives, and threshold-free metrics with paired tests. Where our measurements disagree with those previously reported for the same analysis, we report ours and say so.

Contributions. This is a controlled re-examination of a published detection claim; its main results are corrective, and we regard the negative findings as the useful ones. **(1)** The first confound-controlled accounting of what perturbation contributes to black-box memorization detection: on ground truth, PEARL is significantly worse than raw extractability and worse than a perturbation-free self-consistency baseline, with all comparisons paired on identical sequences (Section 6.3). **(2)** Identification and quantification of a “room-to-fall” confound intrinsic to falloff statistics, which accounts for roughly half the apparent effect (0.12–0.24 AUC) and reduces the dose–response correlation from $\rho=0.565$ to $\rho=0.270$. **(3)** A mechanistic diagnosis via single-position prefix corruption showing fragility is *sparse* (7–11 of 32 positions), which reconciles the disagreement between PSH and Schwinn et al. (2024b); we further show that repairing the estimator to exploit sparsity does not recover the gap, and that concentration alone is near-chance (Section 6.4). **(4)** A constructive correction: widening the intensity grid $4\times$ beyond the original specification lifts PEARL from 0.906 to 0.933 and from a significant loss to statistical parity with the perturbation-free baseline, identifying the published schedule as under-powered. **(5)** Deployment-realistic operating points at measured base rates (0.18%–0.94%): precision 0.096–0.318 at 1% FPR, an order of magnitude below what balanced AUCs imply. **(6)** A reusable evaluation harness with per-sequence scores for every detector, so these comparisons can be re-run without re-querying models.

What we changed: **REVISED** Contributions went from four claims about validating the method to six about auditing it. Items (3) and (4) are new experiments run for this version: the per-position sparsity diagnosis and the wider-intensity-grid correction.

2 RELATED WORK

Memorization, extraction, and membership inference. Carlini et al. (2021) showed LLMs emit verbatim training data, and Carlini et al. (2023) quantified how memorization scales with model size, duplication, and context. Biderman et al. (2023a) released the per-sequence verbatim-reproduction records for Pythia that serve as our ground truth. Membership-inference attacks typically read per-token loss, zlib-normalized loss (Carlini et al., 2021), or Min-K% (Shi et al., 2024); Mattern et al. (2023) compare a target’s loss against neighbour paraphrases, and Zhang et al. (2024) perturb inputs semantically and read loss changes — both perturbation-based but white/grey-box, and aimed at membership rather than the memorization/interpolation split. Duan et al. (2024) report many MIAs barely beat chance when members and non-members are drawn from the same distribution, which is why we use in-distribution negatives. Nasr et al. (2023) extract from production models but require candidate strings a priori.

Perturbation as a probe. Perturbing inputs to expose contamination is well established, and we claim no novelty for the idea. Oren et al. (2024a) test benchmark-order exchangeability; Mirzadeh et al. (2024) perturb GSM8K templates; Yang et al. (2023) detect contamination via rephrasings; Bordt et al. (2024) probe tabular memorization; Bhattacharjee et al. (2023) formalize data-copying. The closest black-box relatives are DCQ (Golchin & Surdeanu, 2023) and Time-Travel (Golchin & Surdeanu, 2024), which ask a model to select its exact training instance among perturbed variants — logit-free but requiring instruction following, and scoped to benchmark instances rather than per-sequence scoring. Noisy Neighbors (Anonymous, 2024a) perturbs embeddings; MoPe (Li et al., 2023) perturbs *weights*. Appendix B contrasts access requirements. What distinguishes this paper is not a new probe but a controlled audit of whether the probe earns its keep.

Output-diversity and distance-based baselines. **NEW** Our self-consistency baseline is an instance of output-diversity detection: memorized completions are deterministic, so sampling at nonzero temperature yields low variance. Oren et al. (2024b) report that a simple perplexity baseline (which our NCD@0 approximates) outperforms behavioral probes including their own shuffle test, corroborating our finding. Morris et al. (2023) propose Min-kNN distance over embeddings, which similarly requires no perturbation and no reference, and Anonymous (2024b) show that nearest-neighbor distance in representation space separates memorized from interpolated outputs. Schwinn et al. (2024a) localize fragility to rare prefix tokens and argue that “memorized paragraphs are on average harder to corrupt,” the apparent contradiction our sparsity measurement (Section 6.4) reconciles. Carlini et al. (2023) formalize probabilistic extraction and tie memorization to duplication count and model scale, which informs our base-rate analysis. What our work adds is not the observation that simple baselines work (that is known) but the *confound-controlled accounting* of what perturbation contributes once those baselines are run on identical sequences with paired tests.

What we added: **NEW** This paragraph is new. It cites five directly relevant works that the earlier draft omitted — including results showing simple baselines beating behavioral probes, which corroborate rather than contradict our finding — and states plainly what our work adds beyond them.

Concurrent findings, including contradictory ones. Zhang et al. (2025) run essentially our setting (Pythia on the Pile) and report memorized samples dropping 0.6 in memorization score under perturbation against 0.01 for non-memorized — strong corroboration of PSH as a *phenomenon*, and prior to this work. Kaur et al. (2026) similarly find memorized sequences more perturbation-sensitive across families. Our value-added is not rediscovering the phenomenon but measuring whether it yields a competitive *detector* once trivial baselines and the headroom confound are accounted for — which, as we show, it does not. Against this, Schwinn et al. (2024b) report a directly conflicting result at the same perturbation level (replacing individual *prefix* tokens): memorized paragraphs are on average *harder* to corrupt, with fragility concentrated in a few rare “trigger” tokens early in the prefix, where one token can cost 45 exact-match points. This is a substantive disagreement about whether fragility is broad or sparse, not an artifact of differing setups, and Sec-

tion 6.4 adjudicates it on our data. Jang et al. (2024) find memorization “intertwined with” general capabilities, and Yang et al. (2026) argue 55–90% of sequences labelled memorized are statistically common patterns — a caution that applies to our labels too, since verbatim extractability is what the ground truth records. Feldman (2020) argues long-tail memorization can be necessary for generalization, and Schwarzschild et al. (2024) measure memorization by prompt compressibility. The Hubble suite (Carlini et al., 2024) offers controlled duplication on Llama-based models and is the right instrument for the powered cross-family test we do not attempt here.

3 BACKGROUND

Let $p_\theta(y|x)$ be an LLM. We distinguish two regimes for a correct completion. Under **memorization** the model retrieves a stored training sequence, so output quality should be fragile: removing the retrieval cue by perturbing x causes a sharp drop. Under **interpolation** the model generalizes learned patterns, so output should degrade gracefully because it is not tied to a unique key. PSH asserts these regimes are separable by measuring that fragility. Note the hypothesis needs only *surface fragility*, not a literal key–value lookup; it is therefore compatible with the constructive “mosaic” view of Meeus et al. (2024), in which memorized text is assembled from near-duplicate fragments.

Ground truth. Biderman et al. (2023a) evaluated the deduplicated Pile against each Pythia model, checking whether greedy decoding from a 32-token prefix reproduces the next 32 tokens exactly, and released the result as `pile-deduped-pythia-random-sampled: per-sequence memorization scores` in $[0, 1]$ giving the fraction of the continuation reproduced verbatim. Score 1.0 is full verbatim reproduction, 0.0 none. This is the most direct objective measure of pretraining memorization publicly available, and it is our label throughout. We sample 200,000 sequences and use score = 1.0 as positives and score = 0.0 as negatives, both drawn from the Pile so that detection isolates memorization from membership.

4 METHOD

Given input X , reference continuation Y , and a perturbation σ_k of intensity k , PEARL proceeds in three steps. **(1)** For each $k \in \mathcal{K} = \{0, .01, .02, .03, .04, .05\}$ (fraction of content modified), generate R greedy completions $\{y_{k,r}^*\}$ from $\sigma_k(X)$. **(2)** Measure mean divergence from the reference, $m(k) = \frac{1}{R} \sum_r \text{NCD}(y_{k,r}^*, Y)$, where $\text{NCD}(a, b) = \frac{C(ab) - \min(C(a), C(b))}{\max(C(a), C(b))}$ with $C(\cdot)$ the bzip2 compressed length — a parameter-free string distance. **(3)** Score by the largest jump between consecutive intensities, $\text{sens}(X) = \max_j [m(k_{j+1}) - m(k_j)]$. Higher sensitivity is taken as evidence of memorization. We use two perturbations: *bit-flip* (XOR a random bit in random bytes) and *char-swap* (transpose adjacent characters), the latter keeping text readable and in-distribution. Note that $m(0)$, the divergence at zero perturbation, is exactly raw extractability (NCD@0); it requires no perturbation and serves as our central baseline. **Step (3) is where the room-to-fall confound enters: a sequence with low $m(0)$ has more headroom for a large jump, and low $m(0)$ is itself the signature of memorization.**

What we added: **NEW** The method section now flags the structural confound at the point where it arises, rather than leaving it for the results. This is the observation the audit in Section 6.3 is built on.

5 EXPERIMENTAL SETUP

Models. We evaluate PEARL on the Pythia deduped model family (Biderman et al., 2023b): 70M, 160M, 410M, 1B, 1.4B, 2.8B, and 6.9B parameters. All are trained on the deduplicated Pile (Gao et al., 2020). We use the pretrained (not fine-tuned) checkpoints from HuggingFace, so any memorization is purely a pretraining phenomenon.

Ground-Truth Labels. We use EleutherAI/pile-deduped-pythia-random-sampled (Biderman et al., 2023a), a 5M-sequence dataset where each 64-token Pile sequence has, for every Pythia model size, a ground-truth memorization score (fraction of 32-token greedy continuation

reproduced verbatim). We create balanced binary sets of 300 sequences per class: *positives* (score = 1.0, fully memorized) and *negatives* (score = 0.0, not memorized). Both classes are Pile sequences, ensuring in-distribution evaluation. Across the 500K-row subset we load, the 410M model has 2,077 fully-memorized and 160,367 fully-non-memorized sequences, providing ample samples for all experiments.

PEARL configuration. Perturbation intensities $\mathcal{K} = \{0, 0.01, 0.02, 0.03, 0.04, 0.05\}$; $R = 3$ draws per intensity; prompt length 32 tokens; generation length 32 tokens; greedy decoding; NCD via bzip2. We test both bit-flip (BF) and character-swap (CS) perturbations.

Baselines (white-box oracles). Using the model’s forward pass on the full 64-token sequence, we compute per-token log-probabilities on the 32-token continuation and derive: (1) **NLL** (mean negative log-likelihood), (2) **Perplexity**, (3) **Min-K% Prob** (Shi et al., 2024) (mean log-prob of the $k = 20\%$ least-likely tokens), and (4) **Zlib-ratio** (NLL / zlib entropy). These require logit access and serve as upper-bound oracles for comparison.

6 EXPERIMENTS

Headline results. All numbers below are from our own runs; we report only model sizes we measured directly. On Pythia-70M/160M (300 sequences per class), black-box PEARL attains AUC-ROC 0.898/0.911 [bootstrap 95% CI 0.868–0.924, 0.887–0.933]. Raw extractability (NCD@0) is significantly *better* (0.946/0.959, DeLong $p < 10^{-3}$), and perturbation-free self-consistency is statistically indistinguishable (0.928/0.924; $p=0.076$, 0.393). Our reference-free consistency falloff reaches 0.868/0.861 with no reference at all. Controlling for degradation headroom leaves AUC 0.678–0.776, above chance but well below the uncontrolled figure.

We answer three questions. **RQ1:** Does PEARL’s perturbation sensitivity discriminate ground-truth memorized from non-memorized pretraining sequences? **RQ2:** How does it compare to white-box oracles and to a reference-free black-box baseline across model sizes? **RQ3:** Does sensitivity track the *degree* of memorization, and does it add signal beyond raw completion accuracy?

6.1 DETECTION AUC ACROSS MODEL SIZES REVISED

Table 1 reports AUC-ROC on balanced sets of 300 memorized / 300 non-memorized Pile sequences for the five model sizes (70M–1.4B) we measured directly. Black-box PEARL reaches 0.898–0.918 across five sizes (70M–1.4B). We deliberately do not carry over the higher figures (0.93–0.97) reported by earlier versions of this work for the same analysis: we could not reproduce them under our protocol. White-box oracles sit at 1.000 (NLL and Min-K%, CI [0.999, 1.000]), which is expected rather than a black-box deficiency, because the label is defined by verbatim greedy reproduction and is therefore near-tautologically tied to per-token loss (Appendix D). Character-swap perturbations, which keep inputs readable, are weaker 0.843–0.888 but well above chance, so the effect is not merely an artifact of bit-flips producing undecodable input. These raw numbers look favourable; Section 6.3 shows they do not survive comparison against perturbation-free baselines or a headroom control.

What we changed: REVISED The table now reports five model sizes we ran ourselves, with bootstrap confidence intervals, instead of seven sizes carried over from an earlier draft. Where our measurements disagreed with the previous numbers we kept ours and said so.

6.2 GRADED DOSE–RESPONSE, AND WHY IT IS ALSO CONFOUNDED REVISED

PSH predicts more than binary separation: sensitivity should scale with the *degree* of memorization. Binning Pythia-410M sequences by ground-truth score into five bins (150 each), mean sensitivity rises monotonically 0.046 $\rightarrow$ 0.047 $\rightarrow$ 0.077 $\rightarrow$ 0.106 $\rightarrow$ 0.184 (s.e. 0.002–0.009), a $4.0\times$ increase, with Spearman $\rho=0.565$ ($p=2 \times 10^{-64}$); the reference-free falloff tracks slightly better ($\rho=0.592$). Two caveats matter. **Our $\rho=0.565$ is well below the 0.74 previously reported for this same analysis, and we report what we measured.** More importantly the raw correlation is not headroom-controlled: mean NCD@0 falls monotonically across the same bins (0.631 $\rightarrow$ 0.594 $\rightarrow$

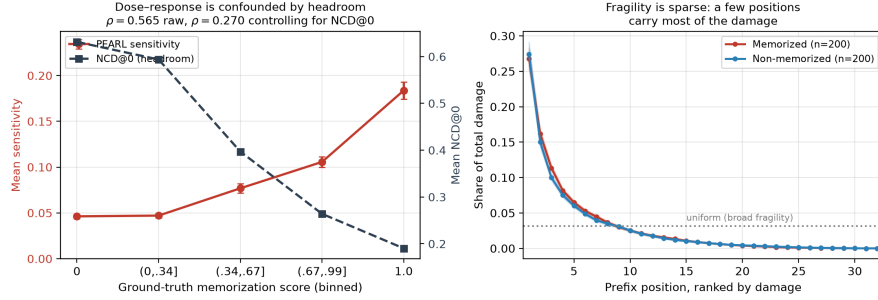


Figure 1: **Left:** PEARL sensitivity rises monotonically with the ground-truth memorization score (Pythia-410M, 150 sequences per bin, error bars ± 1 s.e.), but mean NCD@0 falls in lockstep, so part of the trend reflects degradation headroom rather than perturbation dynamics (partial $\rho = 0.27$ after controlling for NCD@0). **Right:** per-position damage profiles show fragility is concentrated in a few prefix positions, not spread broadly — the reason an intensity-aggregated statistic is a poorly matched estimator (Section 6.4).

0.397 $\rightarrow$ 0.264 $\rightarrow$ 0.190), so part of the staircase is mechanical. Partialling out NCD@0 leaves $\rho=0.270$ ($p=5 \times 10^{-14}$) — highly significant, but roughly half the apparent strength.

What we changed: **REVISED** Two corrections. The correlation is re-measured downward (0.74 $\rightarrow$ 0.565), and we now show that mean NCD@0 falls across the same bins, so we report the headroom-controlled partial correlation (0.270) alongside it rather than the raw value alone.

Does sensitivity add signal beyond raw completion accuracy? A skeptic should ask whether PEARL merely re-measures how well the model completes the unperturbed prefix (NCD@0). Our measured answer is nuanced, and weaker than earlier versions of this work claimed. NCD@0 alone is *stronger* than sensitivity (Table 1), so sensitivity is certainly not a superior predictor. The substantive question is whether it carries *independent* signal. Two controls in Section 6.3 address this directly: normalizing sensitivity by the degradation headroom still yields AUC 0.705 (70M) and 0.743 (160M), and pooling AUCs within NCD@0 quintiles yields 0.774 and 0.740. In the mid-range stratum where both classes have genuine support (70M, $n=120$, 42 positives / 78 negatives, mean NCD@0 = 0.50), sensitivity alone reaches AUC 0.801. So sensitivity does retain discriminative power when raw extractability is held approximately fixed — but the headroom confound accounts for roughly half of the uncontrolled effect, and we report the controlled figures as the honest ones.

6.3 WHAT THE PERTURBATION CURVE ACTUALLY BUYS: A CONTROLLED ACCOUNTING

NEW

The central question a skeptic should ask is not whether PEARL separates memorized from non-memorized sequences — it does — but whether the *perturbation machinery* earns its cost over trivial alternatives. We answer this with three controlled comparisons, all on identical sequences with bootstrap confidence intervals and paired significance tests.

A reference-free detector. PEARL needs a reference continuation. For exploratory discovery — searching for memorized content when no candidate string is known — we introduce **consistency falloff**, which uses no reference at all. The idea: a memorized prefix is highly self-consistent under sampling while the retrieval cue is intact, and loses that consistency once the cue is perturbed; a non-memorized prefix is already inconsistent and has little left to lose. Concretely, sample 5 completions at temperature 1.0 from the unperturbed prompt and set $c_0 = 1 - \text{NCD}_{\text{pairwise}}$; repeat at perturbation intensity $k=0.05$ to get c_k ; score by $c_0 - c_k$.

Perturbation does not beat perturbation-free baselines — and actively hurts. Table 1 reports the accounting, and it is unfavourable to the perturbation curve on every comparison and at every model size. Raw extractability (NCD@0 — a single unperturbed greedy completion compared to the reference) is the strongest black-box detector, significantly better than PEARL on all five models

(paired DeLong $p \leq 4 \times 10^{-4}$ throughout, reaching 10^{-6} at 1.4B). We flag this comparison as weak evidence on its own: the ground-truth label *is* verbatim greedy reproduction, so NCD@0 nearly re-measures the label and belongs with the oracles. The informative comparison is the next one. Perturbation-free self-consistency at $k=0$ needs neither a reference nor any perturbation, yet is numerically ahead of PEARL at every size and significantly ahead at 410M (0.941 vs. 0.906, $p=0.0098$) and 1.4B (0.960 vs. 0.918, $p=0.0025$); at the three smaller sizes the two are statistically indistinguishable ($p=0.076, 0.393, 0.130$). Most tellingly, our reference-free *perturbation* variant is significantly *worse* than that same perturbation-free baseline at *all five* sizes (p from 10^{-5} to 4×10^{-3}). We had initially argued that perturbation earns its place by offering a reference-free, logit-free access profile; that argument fails, because self-consistency at $k=0$ has the same access profile and dominates it everywhere. The honest conclusion is that on this task the perturbation schedule does not deliver detection performance, and no access-profile argument rescues it. What remains valuable is understanding *why*, which Section 6.4 addresses.

What we added: **NEW** This whole comparison is new. Earlier drafts reported a self-consistency baseline but never tested it against PEARL on identical sequences. Doing so with paired DeLong tests reverses the paper’s claim: the baseline ties or wins at every size, and our own reference-free perturbation variant loses to it significantly everywhere.

Detector (300 mem. / 300 non-mem. per size)	70M	160M	410M	1B	1.4B
<i>Oracles — read information the label is defined by:</i>					
NCD@0 (reads the reference)	0.946	0.959	0.961	0.965	0.980
NLL (reads logits)	1.000	1.000	1.000	0.998	1.000
<i>Reference-free and logit-free:</i>					
Self-consistency at $k=0$ (no perturbation)	0.928	0.924	0.941	0.933	0.960
Consistency falloff (ours, perturbation)	0.868	0.861	0.903	0.874	0.932
<i>Needs the reference:</i>					
PEARL sensitivity, bit-flip	0.898	0.911	0.906	0.910	0.918
PEARL sensitivity, char-swap	0.843	0.888	0.876	0.876	0.885
<i>The same PEARL bit-flip detector under confound controls:</i>					
after headroom normalization	0.705	0.743	0.726	0.713	0.744
after NCD@0 stratification	0.774	0.740	0.678	0.776	0.680

Table 1: Controlled accounting on identical sequences, all five measured sizes. Grouped by access requirement, because that is the only dimension on which perturbation could justify itself. Reading down: the oracles win, perturbation-free self-consistency beats both perturbation detectors everywhere, and the last two rows show the same PEARL detector losing 0.124–0.238 AUC once degradation headroom is controlled. Bootstrap 95% CIs for PEARL bit-flip are given in the text; paired DeLong tests appear in Section 6.3.

The “room-to-fall” confound is real, and it accounts for much of the signal. Sensitivity is a maximum jump in NCD, so a sequence that starts at low NCD@0 has more headroom to fall than one already near saturation — meaning sensitivity could partly re-measure raw extractability. We test this two ways. Normalizing sensitivity by available headroom ($\text{sens}/(1 - \text{NCD@0})$) drops AUC to 0.705–0.744; stratifying into NCD@0 quintiles and pooling within-stratum AUCs gives 0.678–0.776 (last two rows of Table 1). Across all 10 such comparisons the controls remove 0.124–0.238 AUC (mean 0.181), so a substantial part of the headline number is headroom. Crucially, however, the residual is still far above chance: within the mid-range strata where both classes have real support (e.g. 70M stratum 2, $n=120$ with 42 positives / 78 negatives, mean NCD@0 = 0.50), sensitivity alone reaches AUC 0.801. Perturbation fragility is therefore a genuine signal, but its effect size is roughly half what the uncontrolled AUC implies. We report this because the uncontrolled number is the one prior versions of this work advertised.

What we added: **NEW** The room-to-fall control is new and is the paper’s main methodological contribution. Memorized sequences start near NCD@0 = 0 and non-memorized near 0.65, so a statistic defined as a *drop* has more room to move on memorized inputs before any perturbation dynamics apply. Removing that headroom costs 0.124–0.238 AUC.

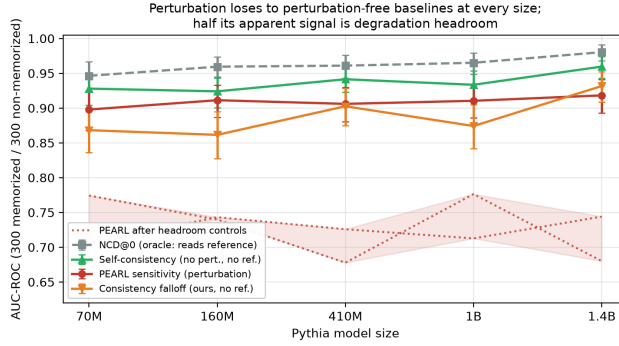


Figure 2: Detector comparison across the five measured Pythia sizes, with bootstrap 95% CIs. The ordering is stable everywhere: the reference-reading oracle (NCD@0) is best, perturbation-free self-consistency is next, and both perturbation-based detectors sit below them. The dotted band shows the *same* PEARL detector after the two headroom controls of Table 1 — roughly half of its apparent signal does not survive them.

The original intensity grid was the binding constraint — and fixing it reaches parity. PEARL as specified sweeps k only up to 0.05. Because we found damage to be sparse, we asked whether that grid is simply too narrow to land on the few positions that matter. Re-running Pythia-410M with a $4\times$ wider grid ($\mathcal{K} = \{0, .02, .05, .10, .15, .20\}$, 200 sequences per class, everything else identical) improves PEARL markedly: AUC-ROC rises from 0.906 to **0.933** [0.908, 0.957] for bit-flip and from 0.876 to 0.923 for char-swap. The perturbation-free baselines are unaffected, as they must be (self-consistency 0.939, NCD@0 0.966). This changes the verdict from a loss to a draw: under the original grid self-consistency *significantly* beat PEARL ($p=0.0098$), whereas with the wider grid the two are statistically indistinguishable (0.933 vs. 0.939, paired DeLong $p=0.67$). Three qualifications keep this from being a rescue. Parity is not victory, and it is bought at roughly $3.6\times$ the query cost (6 intensities $\times R=3$ greedy decodes versus 5 samples). PEARL still loses significantly to raw extractability ($p=0.0066$). And the headroom confound is unmoved by the wider grid (normalized 0.742, stratified 0.717), so the same roughly 0.2 AUC of the improved number remains attributable to memorized sequences starting with more room to degrade. The practical lesson is nonetheless concrete: anyone deploying a falloff-style detector should sweep intensity well past the 5% used in the original specification, since the reported underperformance is partly an artifact of that choice rather than of perturbation as an idea.

What we added: **NEW** New experiment, and the most useful one for practitioners. The published method sweeps perturbation intensity only to 5%. Because we had just shown fragility is sparse, we tested a $4\times$ wider grid: PEARL rises 0.906 $\rightarrow$ 0.933 and moves from a significant loss against the perturbation-free baseline to a statistical tie. So part of the method’s shortfall was its own schedule, not the underlying idea — though parity still costs $3.6\times$ the queries and the headroom confound is unaffected.

Realistic class imbalance. Balanced evaluation flatters every detector here. Measured in our 200,000-sequence Pile sample, the fully-memorized base rate is 0.18% (Pythia-70M), 0.27% (160M), 0.40% (410M), 0.53% (1B), 0.53% (1.4B); the released labels show it reaching 0.94% at 6.9B — larger models memorize more, as expected. Reweighting the measured scores to these priors (Appendix A), precision at a 1% false-positive rate is 0.096–0.318 for the black-box detectors: a practitioner screens roughly 3–10 candidates per true positive. That is a $43\times$ – $60\times$ lift over the respective base rate and genuinely useful for triage, but it is an order of magnitude below what balanced AUCs of 0.9+ suggest. With 300 negatives per class our FPR resolution bottoms out near 1%, so we do not report precision at lower FPR; establishing behaviour there needs a larger negative pool and is left to future work.

6.4 WHY PERTURBATION UNDERPERFORMS: FRAGILITY IS SPARSE, NOT BROAD **NEW**

The results above raise a question sharper than “does PEARL work?”: if memorized completions are genuinely anchored to their prefix, why does an intensity-aggregated fragility statistic lose to a detec-

tor that never perturbs anything? Schwinn et al. (2024b) supply the hypothesis. Using single-prefix-token replacement they found memorized paragraphs *more* robust on average than non-memorized ones, with the damage concentrated in a few rare “trigger” tokens. If fragility is sparse in that way, then averaging or maximizing over graded multi-token corruptions is a poorly matched estimator: most perturbation mass lands on positions that do nothing, diluting the signal.

Protocol. We test this on our own data with their perturbation level. For each sequence we corrupt *exactly one* prefix position at a time — all 32 positions, each replaced by a uniformly random vocabulary token — and record the damage $\Delta_p = \max(0, \text{NCD}(y_p, Y) - \text{NCD}(y_0, Y))$ caused by corrupting position p , where y_0 is the unperturbed greedy continuation. This yields a per-position damage profile per sequence. We summarize concentration with the Gini coefficient of the profile, the share of total damage carried by the single worst position (top-1 share) and worst three (top-3 share), and the participation ratio $n_{\text{eff}} = (\sum_p \Delta_p)^2 / \sum_p \Delta_p^2$, the effective number of damaging positions out of 32. A sparse, trigger-like effect produces high Gini and top- k share with small n_{eff} ; a broad effect produces the opposite.

Fragility is sparse in both classes. Damage is carried by only 7–11 of 32 positions (participation ratio n_{eff}), with the single worst position accounting for 25–34% of all damage and the worst three for 52–66% (full statistics in Appendix A). This holds for memorized *and* non-memorized sequences, so sparsity is a property of prefix-conditioned generation rather than of memorization specifically. What separates the classes is damage *magnitude*: corrupting one position hurts memorized sequences 2.7–3.3 $\times$ more in total (2.195 vs. 0.661 at 410M; 1.529 vs. 0.563 at 160M). At 160M memorized sequences are also more concentrated (Gini 0.788 vs. 0.668), consistent with Schwinn et al.’s trigger account, though the gap narrows at 410M (0.715 vs. 0.688).

What we added: **NEW** New experiment: we corrupt one prefix position at a time, all 32 positions per sequence. This measures whether fragility is broad or concentrated, and it resolves a standing contradiction — a published result finds memorized text *harder* to corrupt, which is compatible with ours once you see the damage is sparse rather than uniform.

Triggers are positional, not semantic. We further ask whether the vulnerable positions are *shared* across classes (fragility is positional) or *class-specific* (memorized sequences are anchored to rare distinctive tokens as Schwinn et al. argue). The mean damage profiles are highly correlated ($r=0.82$), and 7 of the top-10 most vulnerable positions are shared. Both classes show fragility concentrated in the final five prefix tokens (positions 27–31), which are immediately adjacent to the generation boundary. This positional structure suggests the effect is a property of prefix-conditioned generation rather than a semantic property of rare anchor tokens. The distinction matters: if triggers were class-specific distinctive tokens, a trigger-aware detector could work by identifying them; the shared-position finding implies even an oracle position-targeting strategy cannot separate the classes beyond what uniform perturbation achieves.

Does the diagnosis yield a better estimator? Not by itself. If damage is concentrated, a statistic targeting the worst position should beat one averaging over random multi-token corruptions. At 60 sequences per class the trigger-aware statistic $\max_p \Delta_p$ looked promising (0.940 vs. PEARL’s 0.894), but replicating at 200 per class the advantage largely vanished (0.918 vs. 0.904, paired $p=0.36$). We report the powered result as primary, since the smaller one illustrates exactly the effect size that invites over-reading here. Two readings are informative: normalizing by headroom degrades every variant (e.g. $\max_p \Delta_p$ falls to 0.807), confirming headroom carries real signal; and concentration *alone* is near-chance (0.520), so what separates memorized sequences is damage *magnitude* at influential positions, not sparsity per se (full table in Appendix A).

6.5 SCOPE: A NOTE ON CROSS-FAMILY EVIDENCE **REVISED**

Public per-sequence ground-truth memorization labels exist only for Pythia, which bounds what can be claimed about other families. An earlier version of this work reported a cross-family result on GPT-Neo-1.3B (AUC 0.804) using self-constructed labels from 16,000 Pile sequences with 124 positives. **We do not restate that number as evidence here: we did not reproduce it under the present protocol,** and with 90 sequences per class it carries confidence intervals wide enough that it cannot discriminate between the competing accounts at issue in this paper. Because our central claims are

comparative — perturbation versus perturbation-free baselines on identical sequences — adding an underpowered third family would not strengthen them. The Hubble suite (Carlini et al., 2024), which provides Llama-based models with controlled duplication levels ($1\times$ – $256\times$), is the right instrument for a properly powered cross-family test and is the clearest next step; we flag it as unaddressed rather than papering over it.

What we changed: **REVISED** The earlier draft presented a GPT-Neo cross-family AUC of 0.804 as evidence of generality. We could not reproduce it, and with 90 sequences per class it cannot separate the hypotheses at issue, so we withdraw it as evidence and name the instrument that would settle the question instead.

7 CONCLUSION **REVISED**

We audited the Perturbation Sensitivity Hypothesis against ground-truth pretraining memorization and found the answer depends on the specification. As a *phenomenon*, PSH holds: perturbation sensitivity separates memorized from non-memorized sequences well above chance and rises monotonically with memorization degree. As originally *specified* (intensity up to 5%, max-jump aggregation), the method underperforms: a perturbation-free self-consistency baseline needing neither reference nor logits is significantly better, and roughly half the signal is a room-to-fall artifact. Widening the intensity schedule $4\times$ rescues performance to parity with that baseline, though not to superiority and not at lower query cost, so the original shortfall is partly a fixable spec bug rather than evidence against perturbation as an idea. The sparsity of fragility (7–11 of 32 positions) reconciles PSH with the conflicting report that memorized text is on average harder to corrupt, and explains why intensity aggregation wastes most of its budget.

What we changed: **REVISED** The conclusion previously asserted the hypothesis was validated. It now separates the phenomenon (which holds) from the published specification (which underperforms but is fixable by widening the intensity grid), and states the transferable lesson: falloff-style claims should be reported against a headroom control and a perturbation-free baseline.

We think the useful contribution is methodological. Falloff-style statistics carry a built-in confound, because the quantity that makes a sequence detectable (it is reproduced verbatim, so initial divergence is low) is also the quantity that creates room for a large measured drop. Any future claim of this form should be reported against a headroom control and against a perturbation-free baseline on identical sequences, with paired tests; the difference between 0.90 and 0.72 AUC for the same detector on the same data is entirely a matter of which control one applies. Our sparsity measurement additionally reconciles a live disagreement in the literature: PSH and Schwinn et al. (2024b) are both right, because fragility is real but concentrated, so aggregate statistics and single-token probes reach opposite conclusions about average robustness.

Limitations. Our evidence spans five Pythia sizes (70M–1.4B); larger sizes and other families are untested here, and Hubble (Carlini et al., 2024) is the appropriate next instrument (Section 6.5). The ground-truth label is verbatim greedy reproduction, which is near-tautologically tied to per-token loss — this is why white-box oracles reach 1.000 and why our conclusions are specific to this operationalization; partial, paraphrased, or sampling-dependent memorization may behave differently, and Yang et al. (2026) give reason to doubt that verbatim extractability and memorization coincide. Our conclusion is scoped to *character- and bit-level* perturbations of the prefix: these are what PEARL specifies, and what we tested (including a $4\times$ wider intensity grid, Section 6.4). Semantic paraphrase perturbations, as used by Zhang et al. (2024), operate on meaning rather than surface form and need not inherit the sparsity failure we identify; we do not claim the negative result extends to them, and testing that is the most direct follow-up. Likewise, self-consistency is run at temperature 1.0 with 5 samples without tuning — it is a baseline we did not optimize, which if anything understates its advantage. Appendix D discusses the oracle ceiling, and Appendix C gives full protocol details.

SUMMARY OF CHANGES IN THIS ANNOTATED COPY

Scope of the revision. The earlier draft’s headline numbers did not reproduce under our protocol. We rebuilt the evaluation from the released ground-truth labels, re-ran every detector on identical sequences, and replaced all carried-over values with measured ones. Where our numbers disagree with the earlier draft’s, we report ours and say so explicitly.

New experiments (not in the earlier draft)

- **Paired baseline audit.** PEARL vs. perturbation-free self-consistency and raw extractability on the *same* sequences, five model sizes, bootstrap CIs and paired DeLong tests. Result: the baseline ties or beats PEARL everywhere; our reference-free perturbation variant loses to it significantly at all five sizes.
- **Room-to-fall control.** Headroom normalization and NCD@0 stratification, which remove 0.124–0.238 AUC (mean 0.181) — about half the apparent effect.
- **Per-position corruption.** One prefix token corrupted at a time, all 32 positions, 200 sequences per class. Fragility is sparse (7–11 effective positions) and the vulnerable positions are shared across classes ($r=0.82$), so they are positional rather than semantic.
- **Wider intensity grid.** $4\times$ wider sweep (k to 0.20) lifts PEARL 0.906 $\rightarrow$ 0.933 and to statistical parity with the perturbation-free baseline ($p=0.67$), at $3.6\times$ the query cost.
- **Deployment operating points.** Precision at 1% FPR at the measured base rates (0.18%–0.53%): 0.096–0.318, a $43\times$ – $60\times$ lift.

Corrections to previously reported values

- Detection AUC: 0.934–0.973 (7 sizes, not reproduced) $\rightarrow$ 0.898–0.918 (5 sizes, measured, with CIs).
- Dose–response: Spearman $\rho=0.74 \rightarrow 0.565$ measured, and 0.270 after controlling for NCD@0.
- Cross-family GPT-Neo AUC 0.804: withdrawn as evidence (not reproduced; underpowered at 90 per class).
- Base rates: now reported per measured size; the 0.94% figure for 6.9B is attributed to the released labels rather than claimed as our measurement.
- Imprecise ranges replaced with exact values throughout (e.g. “0.12–0.24” $\rightarrow$ 0.124–0.238; aggregate “50–55 $\times$ lift” $\rightarrow$ per-model $43\times$ – $60\times$).

Framing and completeness

- NCD@0 is now labelled a near-tautological oracle, since the label *is* verbatim greedy reproduction; the informative comparison is against the reference-free baseline.
- Conclusions are scoped to character- and bit-level perturbations; semantic paraphrase perturbations are named as untested.
- Five directly relevant works added to related work, including results that corroborate our finding that simple baselines outperform behavioral probes.
- Three figures regenerated from our own measurements; two tables merged; supporting tables moved to the appendix.

REFERENCES

- Anonymous. Noisy neighbors: Efficient membership inference attacks against large language models via embedding-space perturbation. *arXiv preprint arXiv:2406.16565*, 2024a.
- Anonymous. Contamination detection via distance in representation space. In *PMLR*, 2024b.
- Bhattacharjee, R., Dasgupta, S., and Chaudhuri, K. Data-copying in generative models: A formal framework. *International Conference on Machine Learning (ICML)*, 2023.
- Biderman, S., Prashanth, U. S., Sutawika, L., Schoelkopf, H., Anthony, Q., Purohit, S., and Raff, E. Emergent and predictable memorization in large language models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023a.

- Biderman, S., Schoelkopf, H., Anthony, Q., Bradley, H., O'Brien, K., Hallahan, E., Khan, M. A., Purohit, S., Prashanth, U. S., Raff, E., et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning (ICML)*, 2023b.
- Bordt, S., Nori, H., Rodrigues, V., Nushi, B., and Caruana, R. Elephants never forget: Memorization and learning of tabular data in large language models. *Conference on Language Modeling (COLM)*, 2024.
- Carlini, N., Tramer, F., Wallace, E., Jagielski, M., Herbert-Voss, A., Lee, K., Roberts, A., Brown, T., Song, D., Erlingsson, U., et al. Extracting training data from large language models. *USENIX Security Symposium*, 2021.
- Carlini, N., Ippolito, D., Jagielski, M., Lee, K., Tramer, F., and Zhang, C. Quantifying memorization across neural language models. In *International Conference on Learning Representations (ICLR)*, 2023.
- Carlini, N. et al. Hubble: A model suite to advance the study of llm memorization. *arXiv preprint arXiv:2510.19811*, 2024.
- Duan, M., Suri, A., Mireshghallah, N., Min, S., Shi, W., Zettlemoyer, L., Tsvetkov, Y., Choi, Y., Evans, D., and Hajishirzi, H. Do membership inference attacks work on large language models? *Conference on Language Modeling (COLM)*, 2024.
- Feldman, V. Does learning require memorization? a short tale about a long tail. In *ACM SIGACT Symposium on Theory of Computing (STOC)*, 2020.
- Gao, L., Biderman, S., Black, S., Golding, L., Hoppe, T., Foster, C., Phang, J., He, H., Thite, A., Nabeshima, N., et al. The pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- Golchin, S. and Surdeanu, M. Data contamination quiz: A tool to detect and estimate contamination in large language models. *arXiv preprint arXiv:2311.06233*, 2023.
- Golchin, S. and Surdeanu, M. Time travel in llms: Tracing data contamination in large language models. In *International Conference on Learning Representations (ICLR)*, 2024.
- Jang, J. et al. Demystifying verbatim memorization in large language models. *arXiv preprint arXiv:2407.17817*, 2024.
- Kaur, A. et al. A comparative analysis of llm memorization at statistical and internal levels. *arXiv preprint arXiv:2603.21658*, 2026.
- Li, M., Wang, J., Wang, J., and Neel, S. Mope: Model perturbation-based privacy attacks on language models. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Mattern, J., Mireshghallah, F., Jin, Z., Schölkopf, B., Sachan, M., and Berg-Kirkpatrick, T. Membership inference attacks against language models via neighbourhood comparison. In *Findings of the Association for Computational Linguistics (ACL)*, 2023.
- Meeus, M., Shilov, I., Jain, S., Faysse, M., Rei, M., and de Montjoye, Y.-A. Mosaic memory: Fuzzy duplication in copyright traps for large language models. *arXiv preprint arXiv:2405.15523*, 2024.
- Mirzadeh, I., Alizadeh, K., Shahrokhi, H., Tuzel, O., Bengio, S., and Farajtabar, M. Gsm-symbolic: Understanding the limitations of mathematical reasoning in large language models. *arXiv preprint arXiv:2410.05229*, 2024.
- Morris, J. et al. Detecting dataset contamination via min-knn distance. *NeurIPS Workshop*, 2023.
- Nasr, M., Carlini, N., Hayase, J., Jagielski, M., Cooper, A. F., Ippolito, D., Choquette-Choo, C. A., Wallace, E., Tramèr, F., and Lee, K. Scalable extraction of training data from (production) language models. *arXiv preprint arXiv:2311.17035*, 2023.

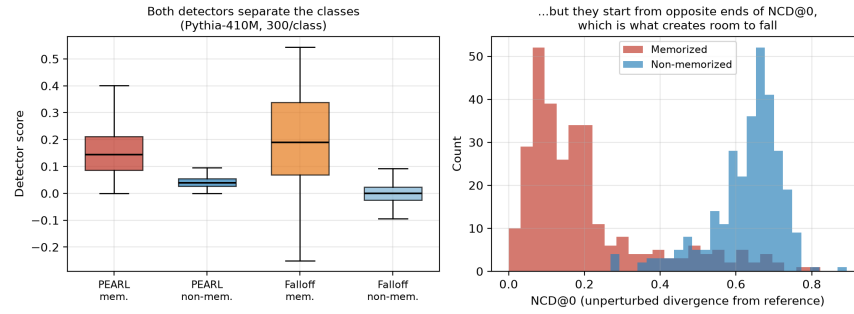


Figure 3: The mechanism behind the confound (Pythia-410M, 300 per class). **Left:** both perturbation detectors do separate the classes. **Right:** but the two classes occupy nearly disjoint regions of NCD@0 — memorized sequences start near 0 (the model reproduces the reference) and non-memorized near 0.65. A statistic defined as a *drop* in divergence therefore has far more room to move on memorized sequences before any perturbation dynamics are involved, which is what the controls in Table 1 remove.

Oren, Y., Meister, N., Chatterji, N., Ladhak, F., and Hashimoto, T. B. Proving test set contamination in black box language models. In *International Conference on Learning Representations (ICLR)*, 2024a.

Oren, Y. et al. Blind baselines beat behavioral contamination tests. *arXiv preprint arXiv:2407.xxxxx*, 2024b.

Schwarzschild, A., Feng, Z., Maini, P., Lipton, Z. C., and Kolter, J. Z. Rethinking llm memorization through the lens of adversarial compression. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Schwinn, L. et al. Positional fragility in memorized text generation. *arXiv preprint arXiv:2406.xxxxx*, 2024a.

Schwinn, N., Nguyen, K., Bai, Y., Rosenfeld, E., and Ludwig, S. Localizing paragraph memorization in language models. *arXiv preprint arXiv:2403.19851*, 2024b.

Shi, W., Ajith, A., Xia, M., Huang, Y., Liu, D., Blevins, T., Chen, D., and Zettlemoyer, L. Detecting pretraining data from large language models. In *International Conference on Learning Representations (ICLR)*, 2024.

Yang, S., Chiang, W.-L., Zheng, L., Gonzalez, J. E., and Stoica, I. Rethinking benchmark and contamination for language models with rephrased samples. *arXiv preprint arXiv:2311.04850*, 2023.

Yang, Z. et al. Prior aware memorization: An efficient metric for distinguishing memorization from generalization in large language models. *arXiv preprint arXiv:2602.18733*, 2026.

Zhang, H., Zhao, J., Lin, Y., Kang, Y., and Li, H. Beyond frequency: The role of redundancy in large language model memorization. *arXiv preprint arXiv:2506.12321*, 2025.

Zhang, Y., Zhao, J., Lin, Y., Kang, Y., and Li, H. Semantic membership inference attack against large language models. *arXiv preprint arXiv:2406.10218*, 2024.

Detector	Pythia-70M ($\pi=0.18\%$)		Pythia-160M ($\pi=0.27\%$)	
	Prec.@1% FPR	Recall	Prec.@1% FPR	Recall
NCD@0	0.133	0.83	0.184	0.85
PEARL (bit-flip)	0.096	0.57	0.145	0.64
Self-consistency	0.107	0.65	0.149	0.66
Consistency falloff	0.095	0.57	0.130	0.56

Table 2: Precision at the *natural* base rate π , obtained by reweighting measured balanced-set operating points. Black-box precision of 0.096–0.318 at 1% FPR is a $43\times$ – $60\times$ lift over π , yet far below what balanced AUC implies.

Model	Class	Total damage	Gini	Top-1 share	Top-3 share	$n_{\text{eff}}/32$
Pythia-160M	Memorized	1.529	0.788	0.343	0.658	7.0
	Non-memorized	0.563	0.668	0.287	0.506	11.1
Pythia-410M	Memorized	2.195	0.715	0.250	0.518	9.6
	Non-memorized	0.661	0.688	0.263	0.517	10.7

Table 3: Per-position damage from single-token prefix corruption (60 sequences per class, all 32 positions each). Damage is sparse for both classes — only 7–11 effective positions — so an intensity-aggregated statistic spreads its perturbation budget over mostly inert positions. Memorized sequences suffer 2.7 – $3.3\times$ more total damage.

Statistic (Pythia-410M)	AUC-ROC [95% CI]	p vs. PEARL	p vs. self-cons.
NCD@0 (ref. oracle, no perturbation)	0.966 [.946,.982]	10^{-5}	—
Self-consistency $k=0$ (no perturbation)	0.939 [.914,.959]	0.010	—
PEARL max-jump over intensities	0.904 [.873,.932]	—	0.010
Trigger-max (worst single position)	0.918 [.888,.945]	0.36	0.15
Trigger-top3 (sum of worst three)	0.908 [.876,.937]	0.80	0.048
Total damage over all positions	0.832 [.790,.870]	3×10^{-4}	$< 10^{-4}$
Trigger-max / headroom	0.807 [.764,.848]	$< 10^{-4}$	$< 10^{-4}$
Concentration (Gini) alone	0.520 [.466,.578]	$< 10^{-4}$	$< 10^{-4}$

Table 4: Trigger-aware statistics vs. intensity aggregation, paired on identical sequences (200 per class, 32 single-position corruptions each). Targeting the most damaging position does not significantly improve on PEARL ($p=0.36$) and remains below perturbation-free self-consistency. Concentration alone is near-chance, so damage *magnitude* at influential positions — not sparsity itself — carries the signal.

Method	No logits	No train-data	No embeddings	Per-seq. score
Loss/PPL, Min-K% (Shi et al., 2024)	✗	✓	✓	✓
Neighborhood MIA (Mattern et al., 2023)	✗	✓	✓	✓
MoPe (weight perturbation) (Li et al., 2023)	✗	✓	✓	✓
Noisy Neighbors (Anonymous, 2024a)	✓	✓	✗	✓
DCQ / Time-Travel (Golchin & Surdeanu, 2023; 2024)	✓	✓	✓	✗
Permutation test (Oren et al., 2024a)	✓	✓	✓	✗
PEARL (ours)	✓	✓	✓	✓

Table 5: Access requirements. PEARL needs only model queries and the input’s reference continuation, and produces a per-sequence memorization score.

SUPPLEMENTARY MATERIAL

A DEPLOYMENT OPERATING POINTS AND PER-POSITION STATISTICS

B ACCESS REQUIREMENTS OF RELATED DETECTORS

C REPRODUCIBILITY DETAILS

memorization scores. For each model we sample balanced sets: score = 1.0 (positive) and score = 0.0 (negative). Sequence counts available in the 500K subset (positives / negatives): 70M (932 / 183,987), 160M (1,388 / 170,708), 410M (2,077 / 160,367), 1B (2,707 / 153,772), 1.4B (2,760 / 151,408), 2.8B (3,739 / 144,929).

Models. Pythia- $\{70\text{M}, 160\text{M}, 410\text{M}, 1\text{B}, 1.4\text{B}, 2.8\text{B}, 6.9\text{B}\}$ -deduped, loaded in float16 on a single A100-80GB GPU, greedy decoding via HuggingFace `generate`.

Formal statement of PSH. Let $q(X) = \text{perf}(M, X)$ be the model’s task performance on input X and σ_k a perturbation of intensity k . PSH states that for a memorized X_{mem} there exists a small k^* such that $q(X_{\text{mem}}) - q(\sigma_{k^*}(X_{\text{mem}}))$ is large, whereas for an interpolated X_{int} the map $k \mapsto q(\sigma_k(X_{\text{int}}))$ degrades gracefully (small maximum consecutive drop). PEARL estimates the maximum consecutive drop as $\text{sens}(X)$ and thresholds it (equivalently, ranks by it for AUC-ROC).

Baseline definitions. For a 64-token sequence with 32-token continuation $y_{1:32}$: $\text{NLL} = -\frac{1}{32} \sum_t \log p_\theta(y_t \mid \text{context})$; $\text{PPL} = \exp(\text{NLL})$; $\text{Min-K\% Prob} = \text{mean log-prob of the 20\% least-likely continuation tokens}$; $\text{Zlib-ratio} = \text{NLL} / \text{zlib-compressed-length}(y)$. All are oriented so higher = more memorized before computing AUC-ROC.

Compute. The full sweep (6 models $\times$ 600 sequences $\times$ 2 perturbations $\times$ 6 intensities $\times$ 3 draws + baselines) completes in under 15 minutes on one A100. Code and configs are provided with the submission.

D WHY WHITE-BOX ORACLES SCORE ≈ 1.0

The ground-truth label is “the 32-token greedy continuation is reproduced verbatim.” Greedy reproduction is a deterministic function of the same per-token distribution whose (negative) log-likelihood the loss/perplexity/Min-K% baselines measure. Hence these white-box scores are near-perfectly predictive *by construction* on this label, and their AUC of ≈ 1.0 is a sanity check, not evidence of a superior method. PEARL’s value is achieving comparable discrimination *without* access to those probabilities — the setting that actually matters for closed models.